BSc (Hons) in Information Technology Specializing in Software Engineering

IT4010 – Research Project

Group - 25-26J-452

Checklist 1 - Part B – Design & Mockups

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Kavinda S.G.D	IT22603654

Component 1 - Intelligent Fabric Capture, Anomaly Pre-Screening, And Quality Analytics Module.

1. Sketches

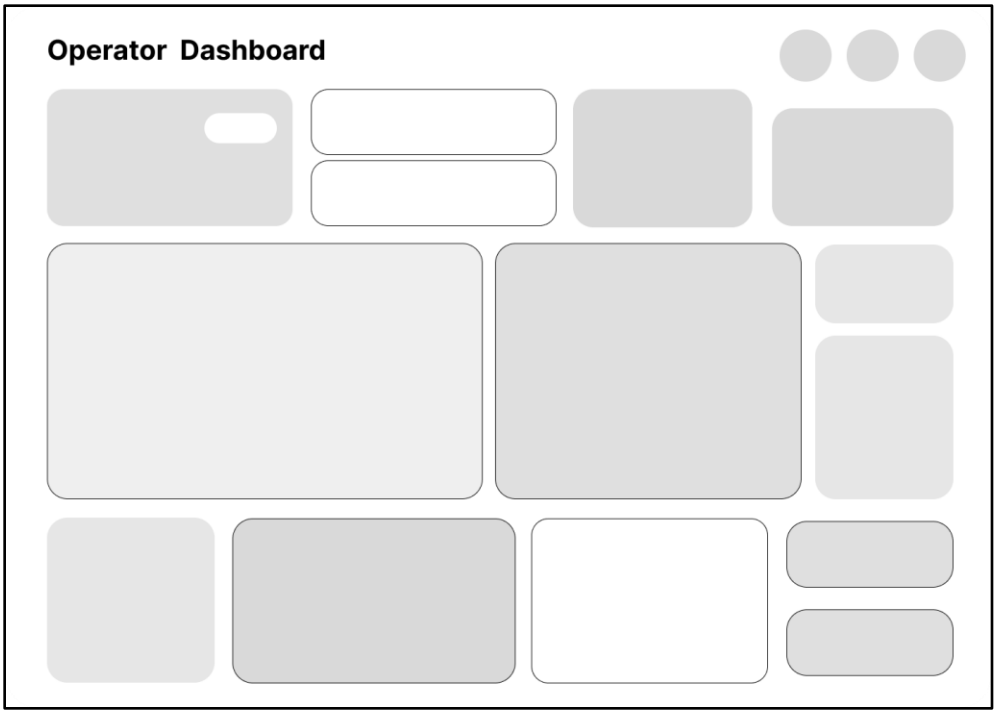


Figure 1: Component 1 Sub-Page 1 – Frame Capture Module Sketch

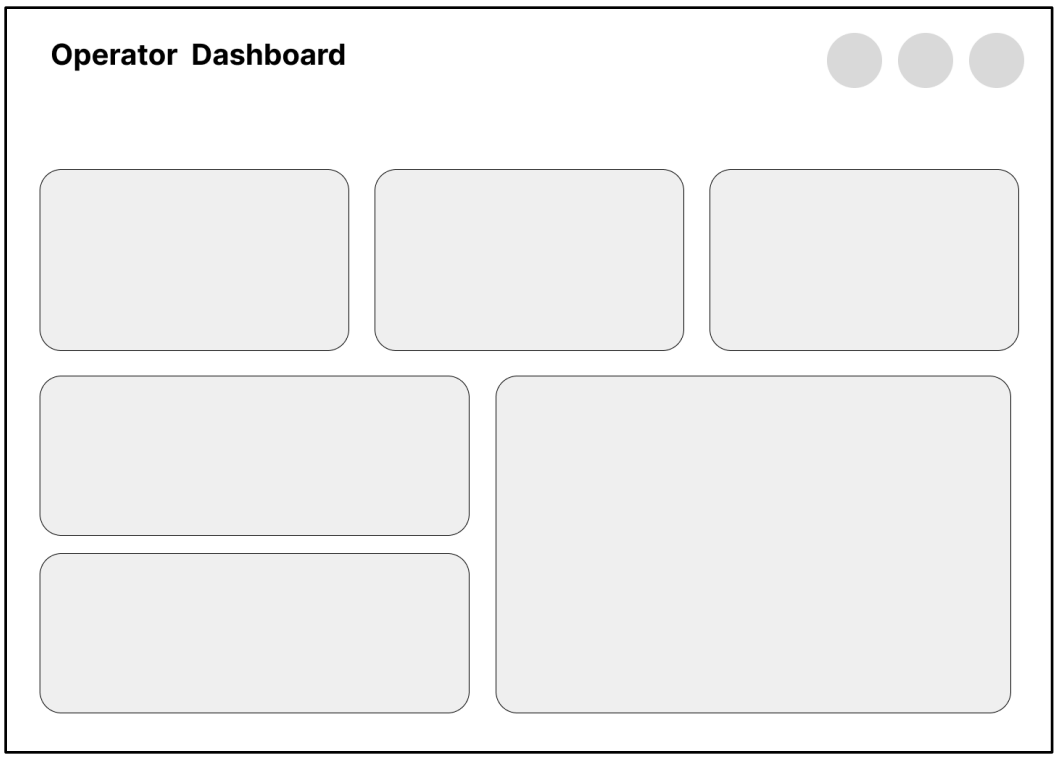


Figure 2: Component 1 Sub-Page 2 – Anomaly-Detection Module Sketch

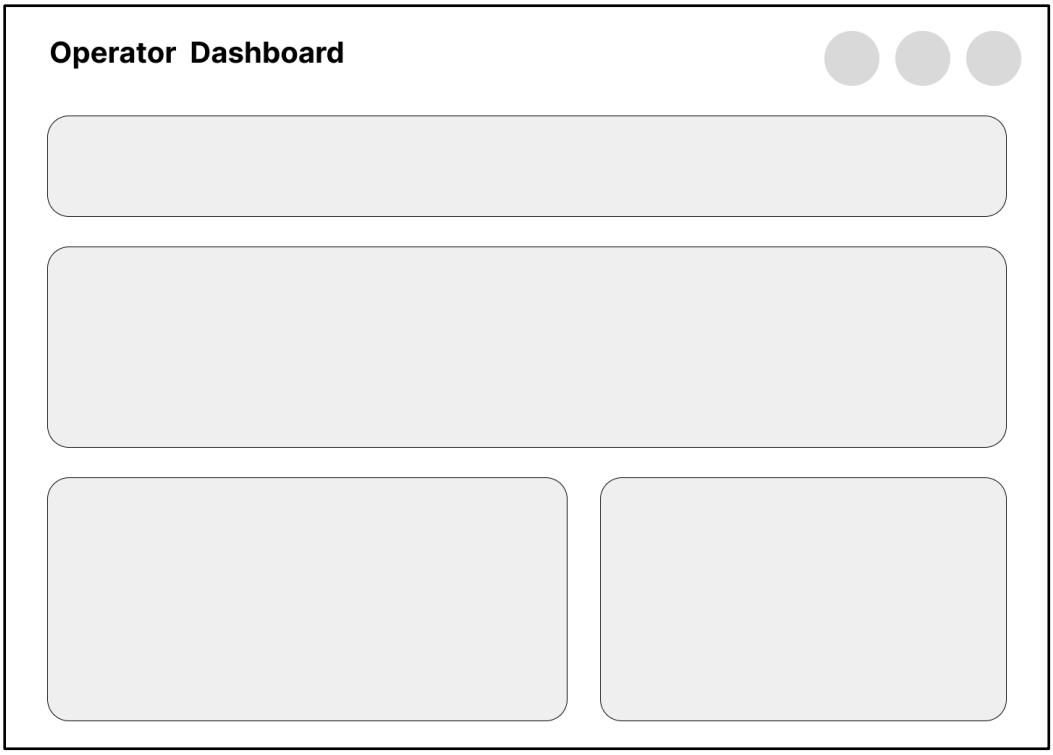


Figure 3: Component 1 Sub-Page 3 – Quality Analysis Module Sketch

2. Mockups

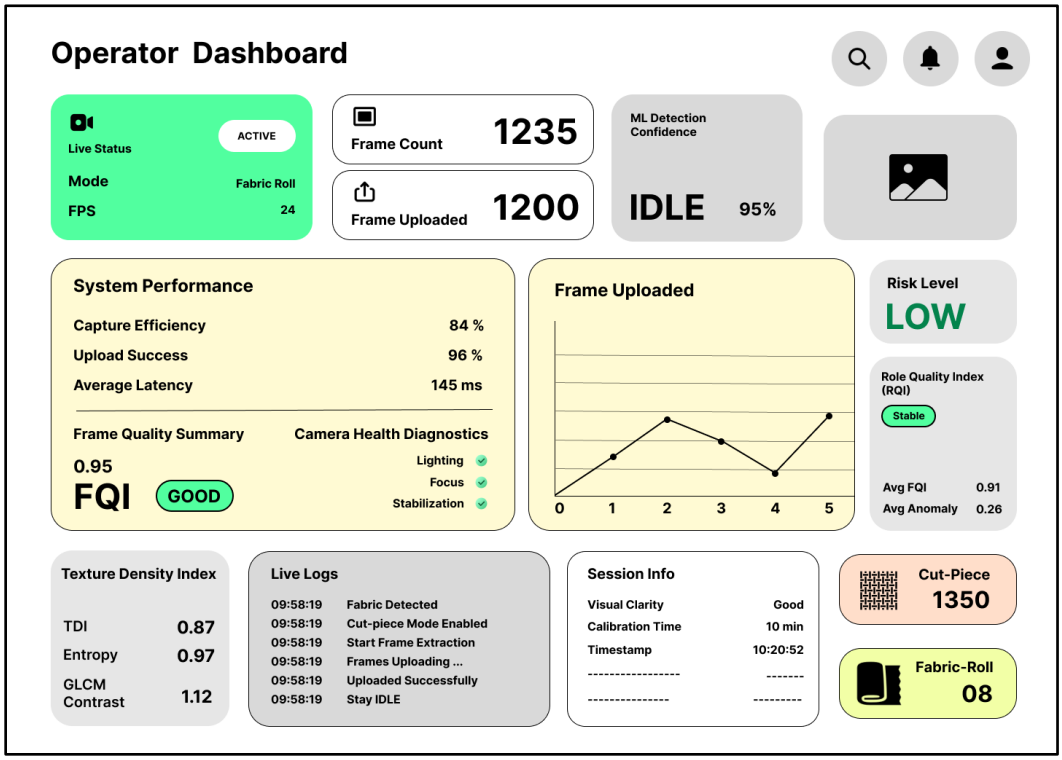


Figure 4: Component 1 Sub-Page 1 - Mockup

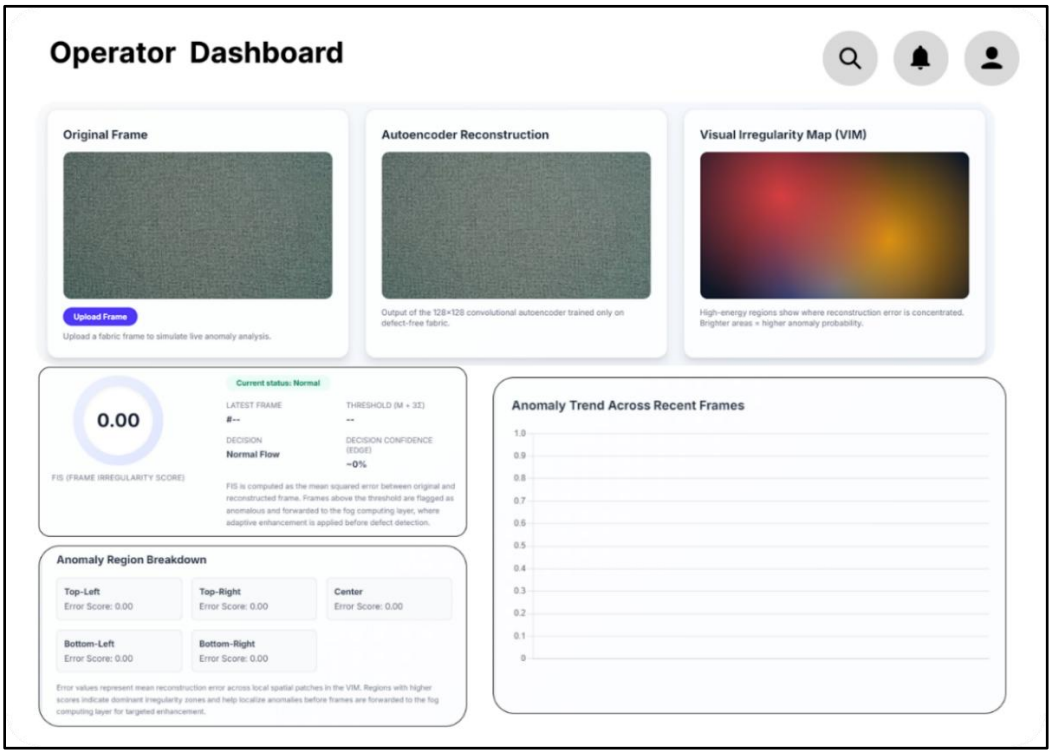


Figure 5: Component 1 Sub-Page 2 - Mockup

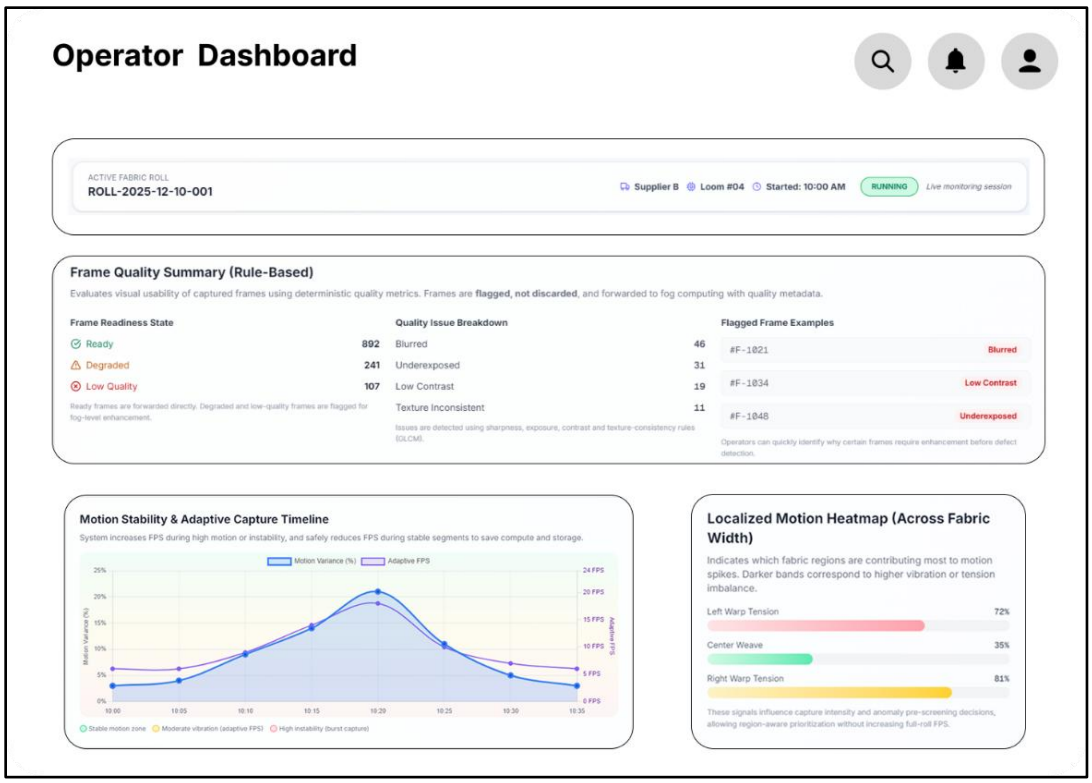


Figure 6: Component 1 Sub-Page 3 - Mockup

3. After Improvements

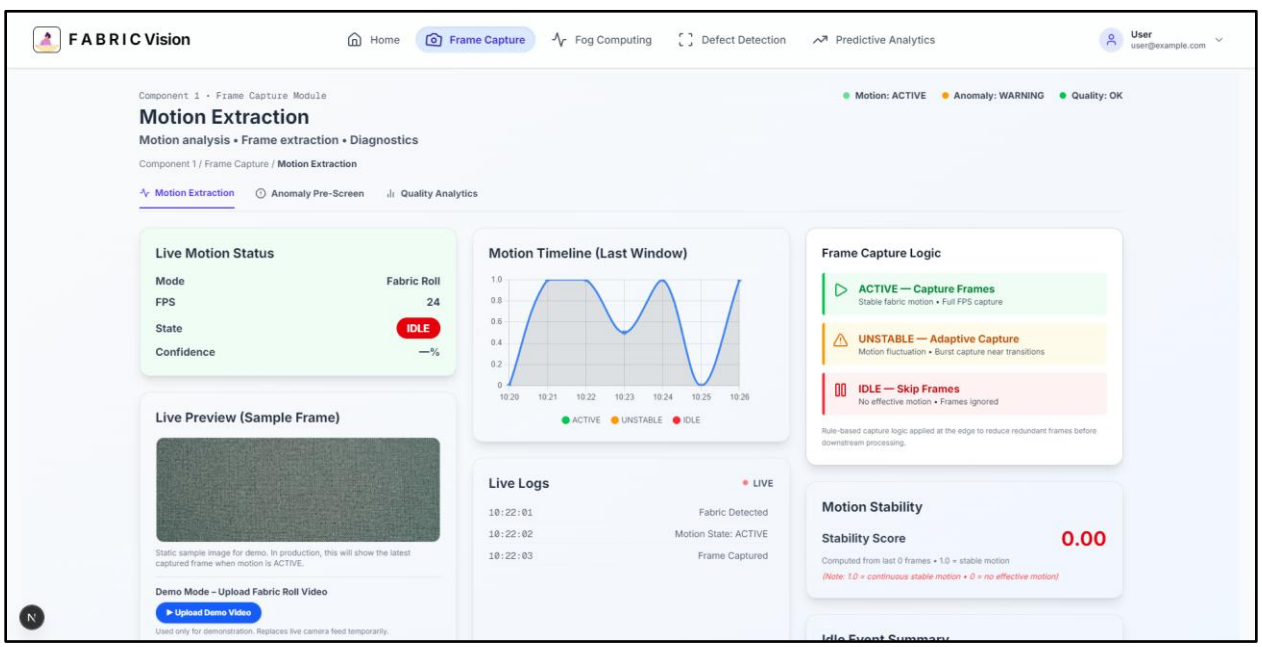


Figure 7: Component 1 Sub-Page 1 – After Improvements

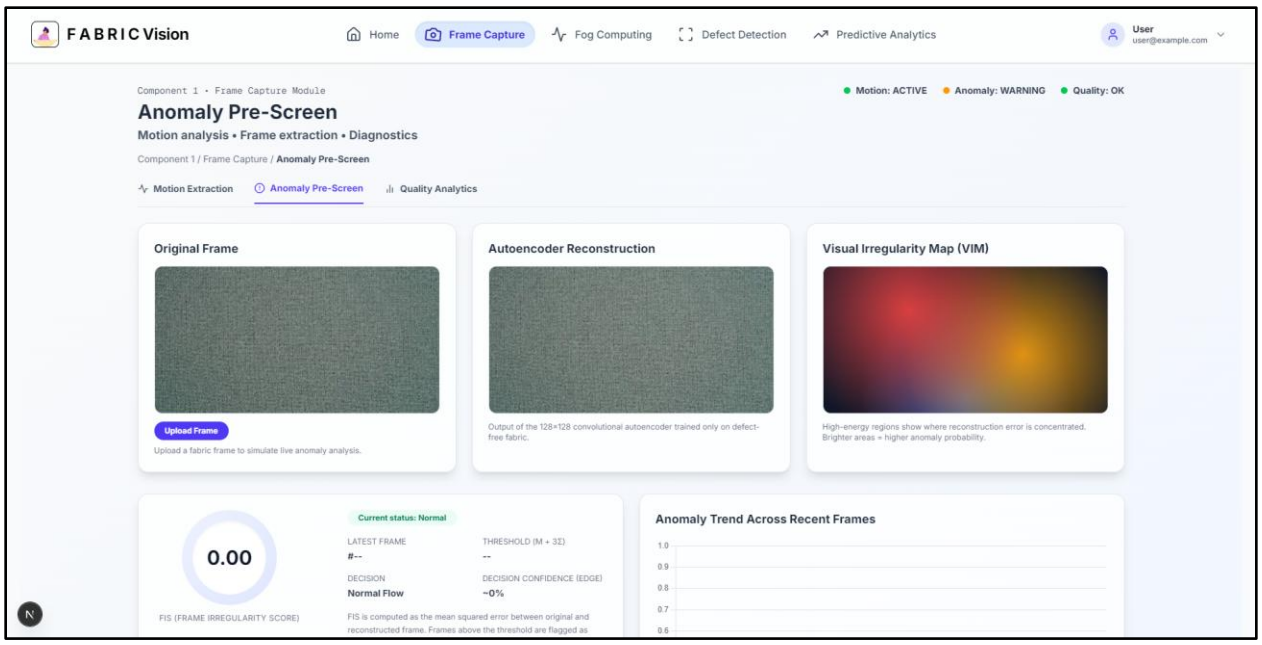


Figure 8: Component 1 Sub-Page 2 – After Improvements

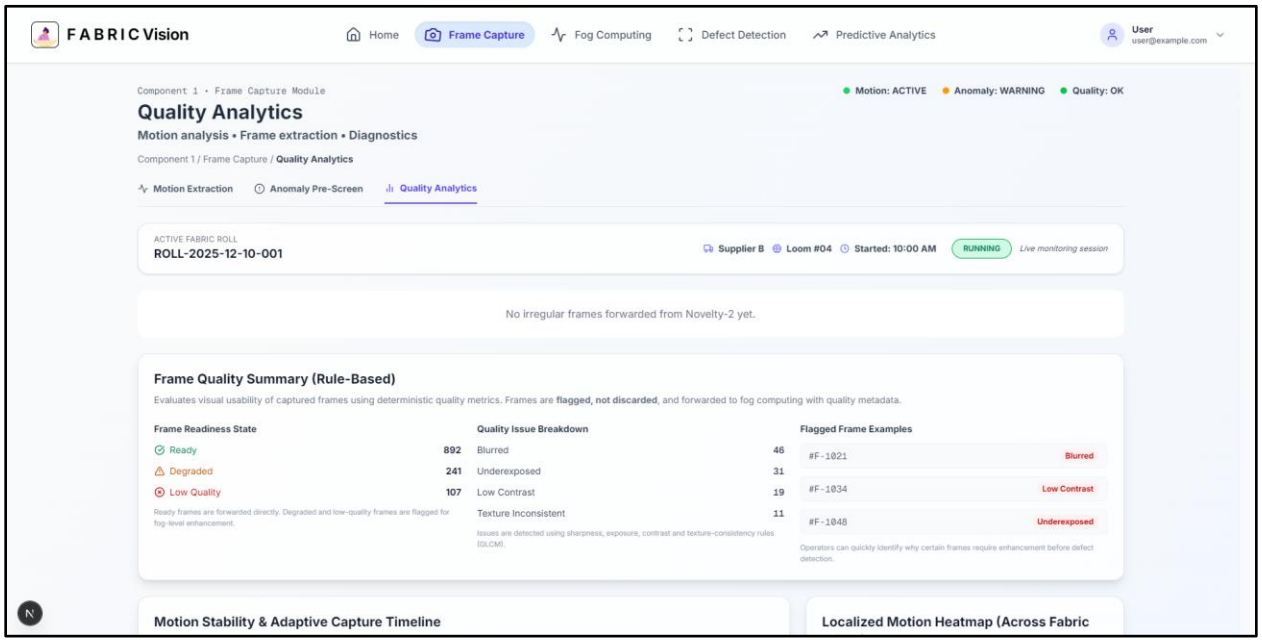
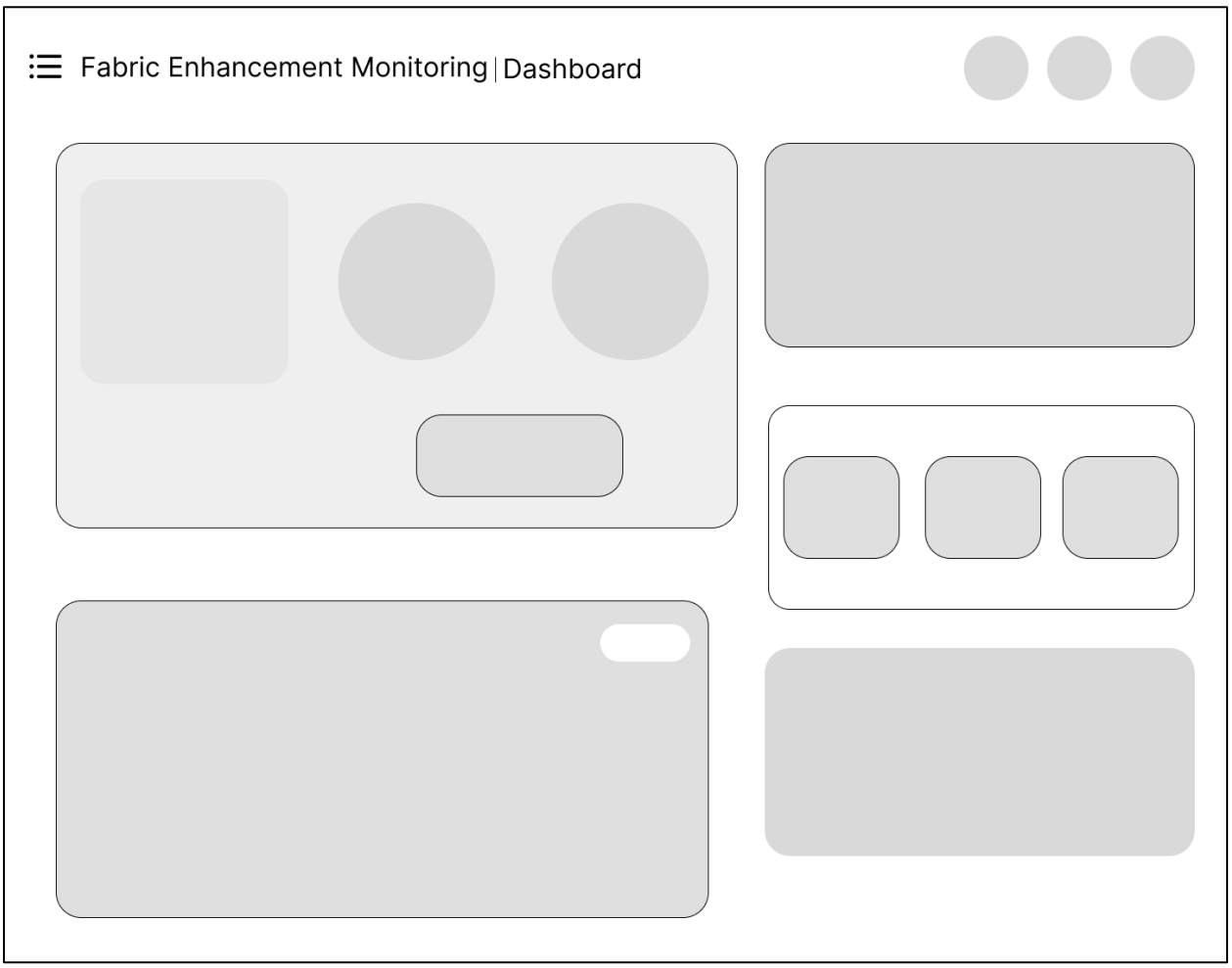
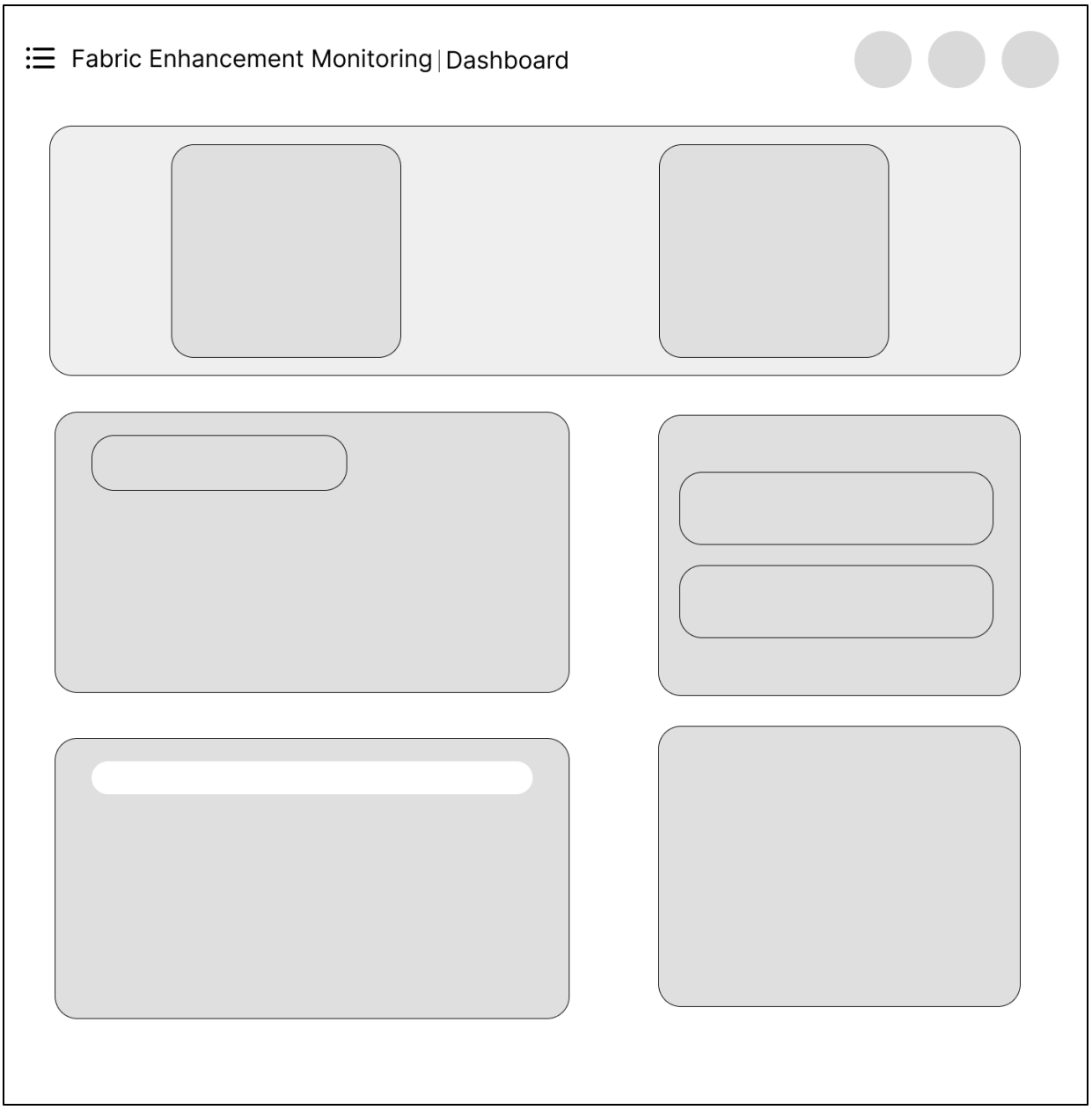


Figure 9: Component 1 Sub-Page 2 – After Improvements

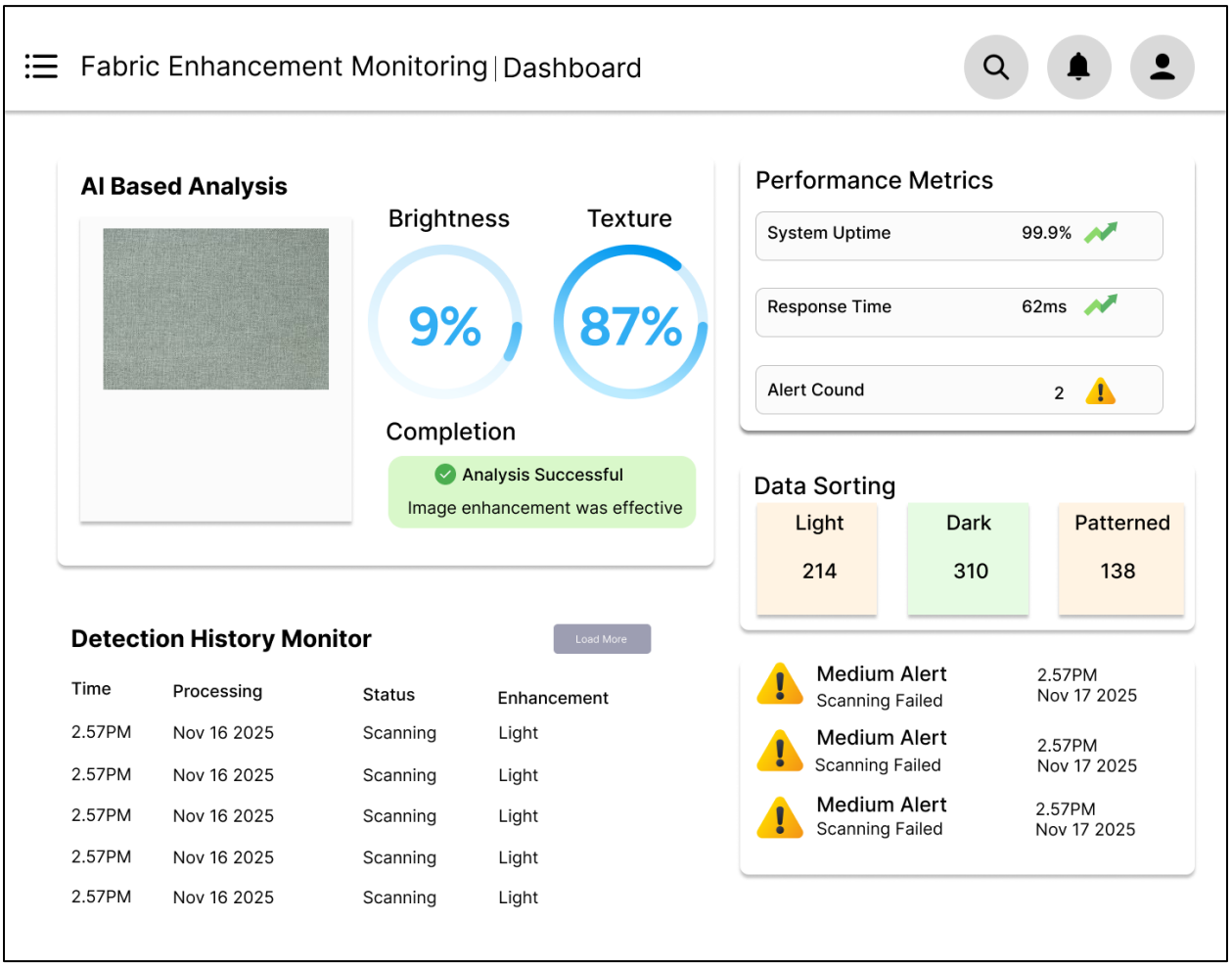
Component 2 - Edge Level Fog Processing & Image Enhancement

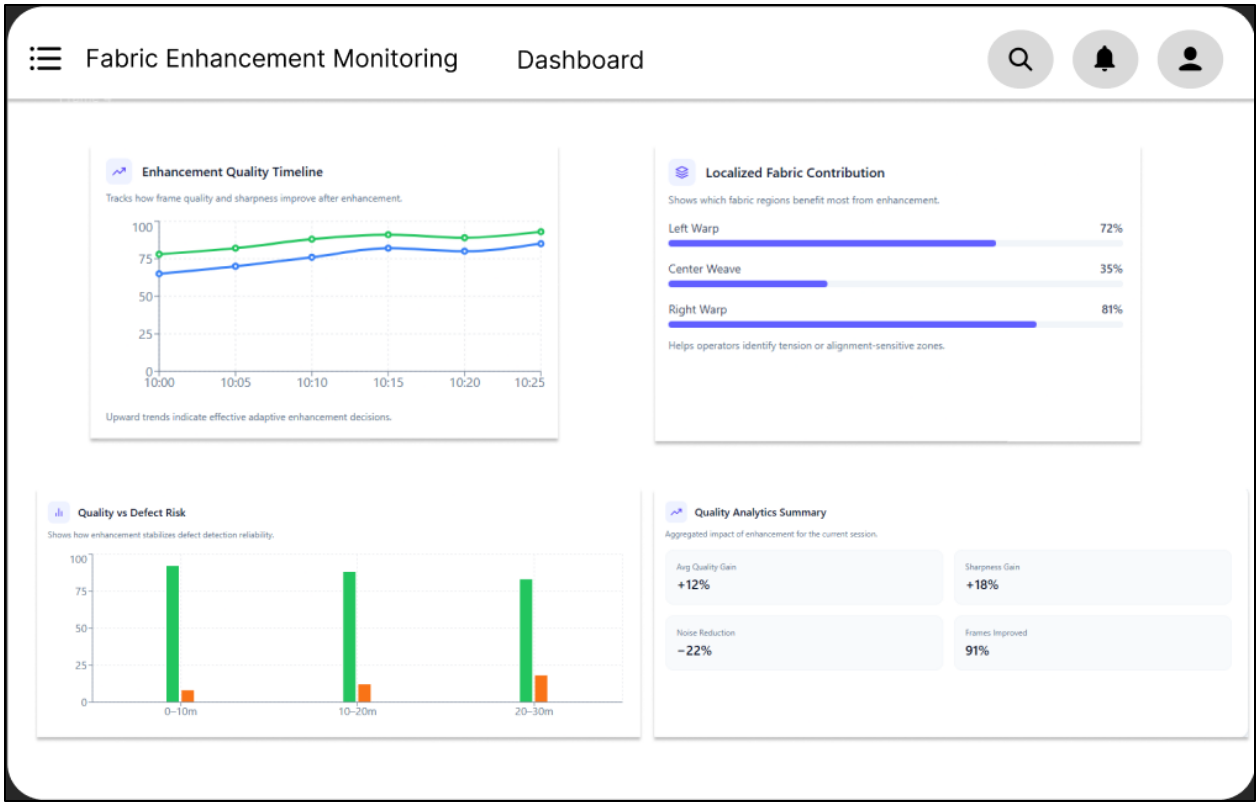
Sketches



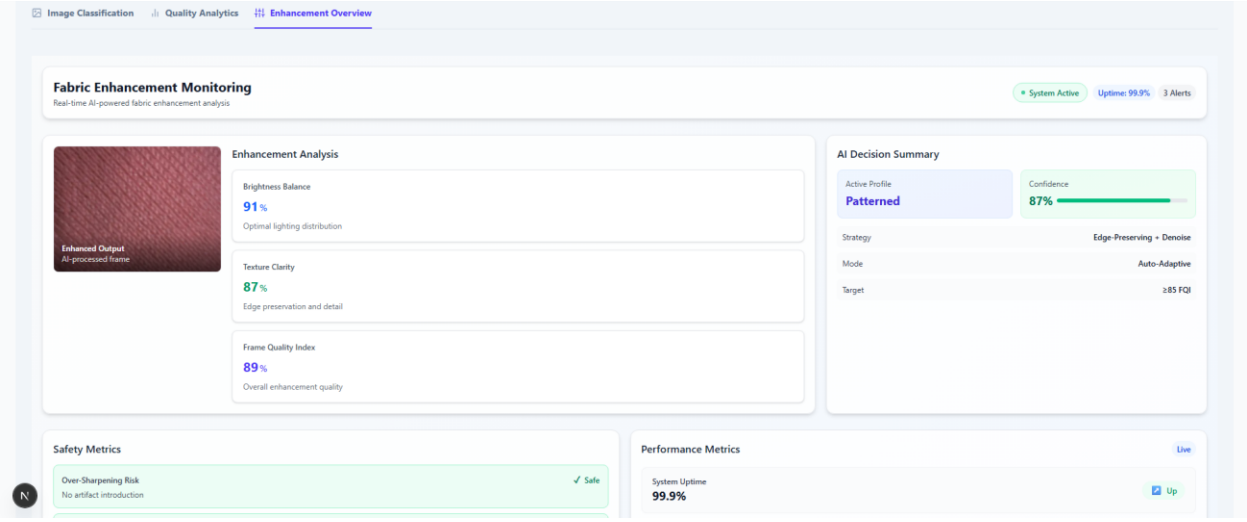

Fabric Enhancement Monitoring | Dashboard

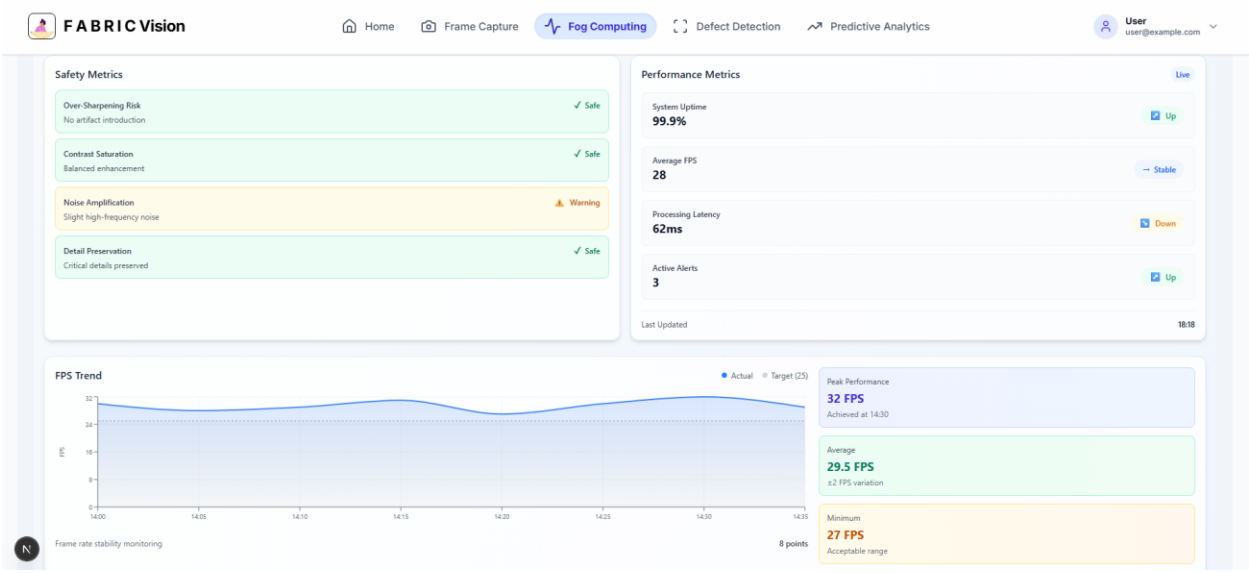
Mockups

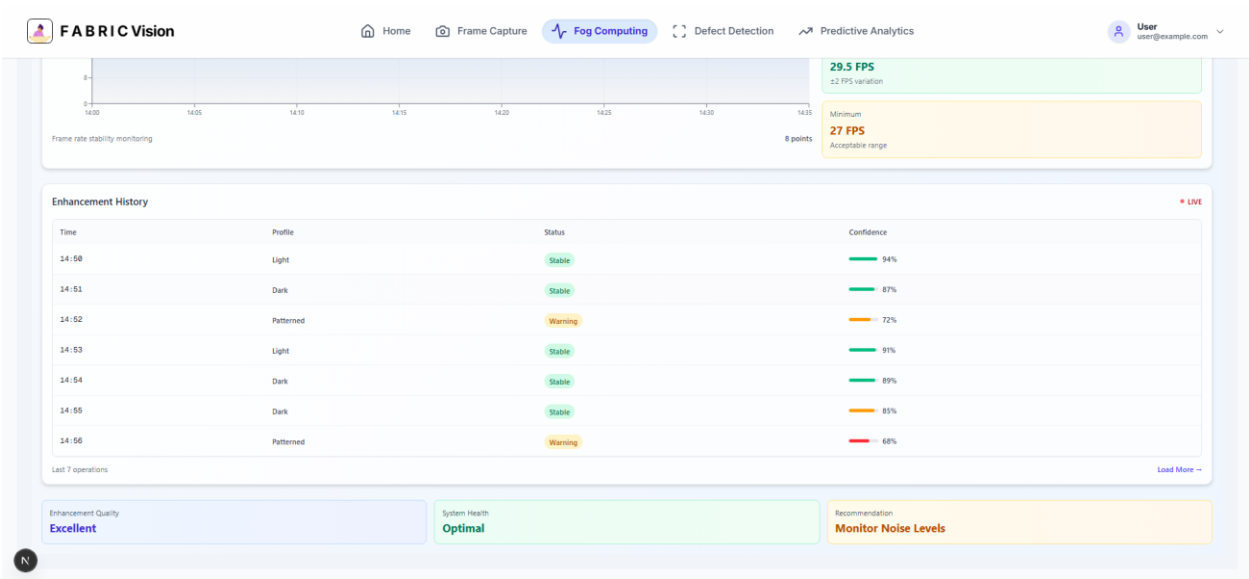

Fabric Enhancement Monitoring | Dashboard
AI Based Analysis
Brightness
9%
Texture
87%
Completion
Analysis Successful
Image enhancement was effective
Performance Metrics
System Uptime
99.9%
Response Time
62ms
Alert Cound
2
Data Sorting
Light
214
Dark
310
Patterned
138
Detection History Monitor
Load More
Time
Processing
Status
Enhancement
2.57PM
Nov 16 2025
Scanning
Light
Medium Alert
Scanning Failed
2.57PM
Nov 17 2025

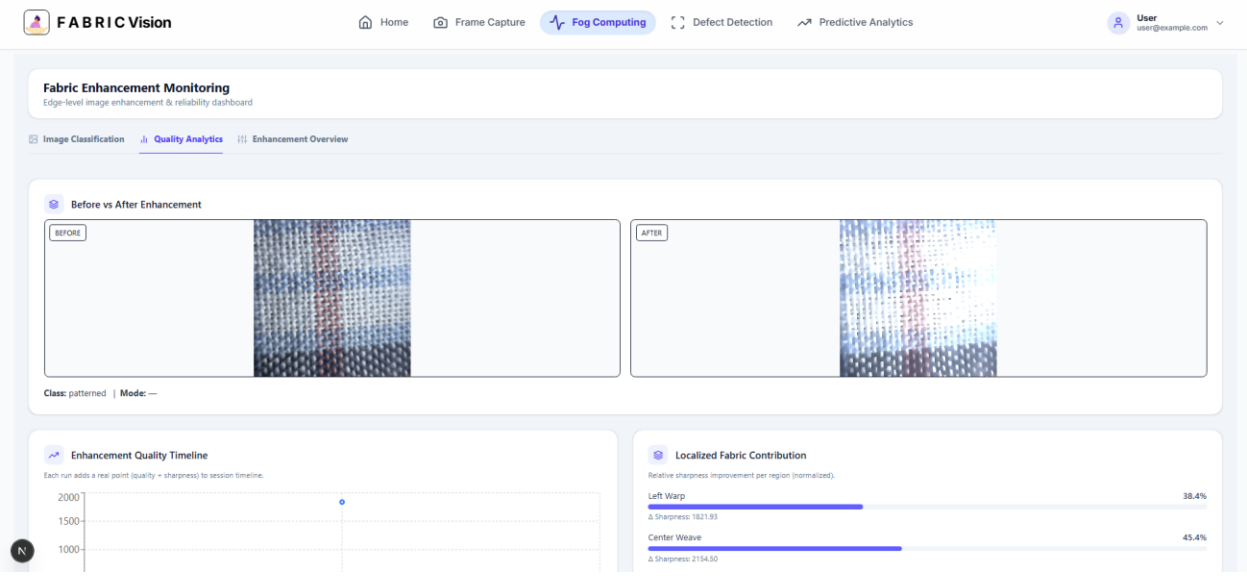


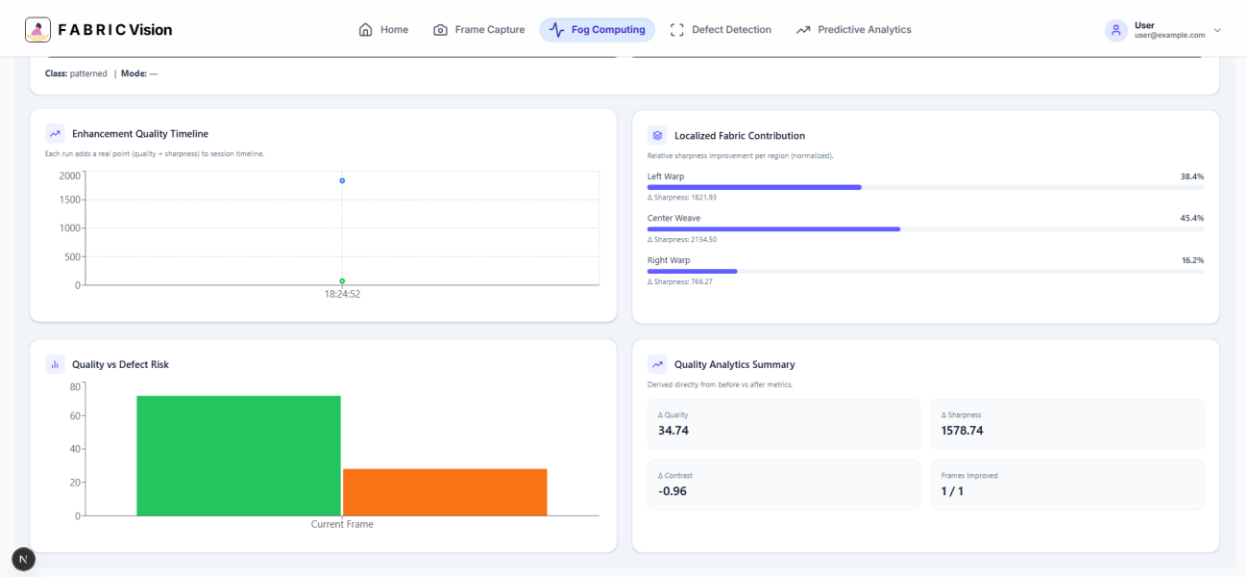
After Improvements




FABRIC Vision
Home
Frame Capture
Fog Computing
Defect Detection
Predictive Analytics
User
user@example.com
Safety Metrics
Over-Sharpening Risk
No artifact introduction
Safe
Contrast Saturation
Balanced enhancement
Safe
Noise Amplification
Slight high-frequency noise
Warning
Detail Preservation
Critical details preserved
Safe
Performance Metrics
Live
System Uptime
99.9%
Up
Average FPS
28
Stable
Processing Latency
62ms
Down
Active Alerts
3
Up
Last Updated
18:18
FPS Trend
Actual
Target (25)
FPS
32
24
16
8
0
14:00
14:05
14:10
14:15
14:20
14:25
14:30
14:35
Frame rate stability monitoring
8 points
Peak Performance
32 FPS
Achieved at 14:30
Average
29.5 FPS
±2 FPS variation
Minimum
27 FPS
Acceptable range


FABRIC Vision
Home
Frame Capture
Fog Computing
Defect Detection
Predictive Analytics
User
user@example.com
29.5 FPS
±2 FPS variation
Minimum
27 FPS
Acceptable range
Frame rate stability monitoring
8 points
Enhancement History
LIVE
Time
Profile
Status
Confidence
14:50
Light
Stable
94%
14:51
Dark
Stable
87%
14:52
Patterned
Warning
72%
14:53
Light
Stable
91%
14:54
Dark
Stable
89%
14:55
Dark
Stable
85%
14:56
Patterned
Warning
68%
Last 7 operations
Load More →
Enhancement Quality
Excellent
System Health
Optimal
Recommendation
Monitor Noise Levels


FABRIC Vision
Home
Frame Capture
Fog Computing
Defect Detection
Predictive Analytics
User
user@example.com
Fabric Enhancement Monitoring
Edge-level image enhancement & reliability dashboard
Image Classification
Quality Analytics
Enhancement Overview
Before vs After Enhancement
BEFORE
AFTER
Class: patterned | Mode: —
Enhancement Quality Timeline
Each run adds a real point (quality + sharpness) to session timeline.
2000
1500
1000
Localized Fabric Contribution
Relative sharpness improvement per region (normalized).
Left Warp
38.4%
Δ Sharpness: 1821.93
Center Weave
45.4%
Δ Sharpness: 2154.50


FABRIC Vision
Home
Frame Capture
Fog Computing
Defect Detection
Predictive Analytics
User
user@example.com
Class: patterned | Mode: —
Enhancement Quality Timeline
Each run adds a real point (quality + sharpness) to session timeline.
2000
1500
1000
500
0
18:24:52
Localized Fabric Contribution
Relative sharpness improvement per region (normalized).
Left Warp
38.4%
Δ Sharpness: 1821.93
Center Weave
45.4%
Δ Sharpness: 2154.50
Right Warp
16.2%
Δ Sharpness: 766.27
Quality vs Defect Risk
80
60
40
20
0
Current Frame
Quality Analytics Summary
Derived directly from before vs after metrics.
Δ Quality
34.74
Δ Sharpness
1578.74
Δ Contrast
-0.96
Frames Improved
1 / 1

Component 3 - Real-time Fabric Defect Detection & Localization Using Cv & Iot-based Rotary Encoder Tracking

Sketches

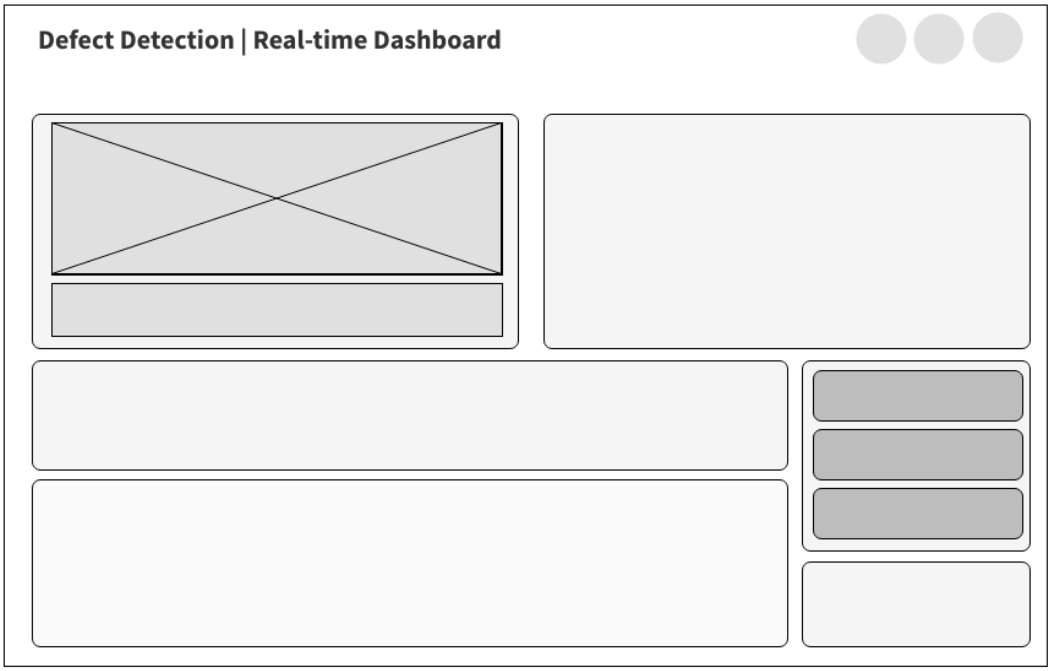


Figure 10: Component 3 Sub-Page 1 – Realtime detection Module Sketch

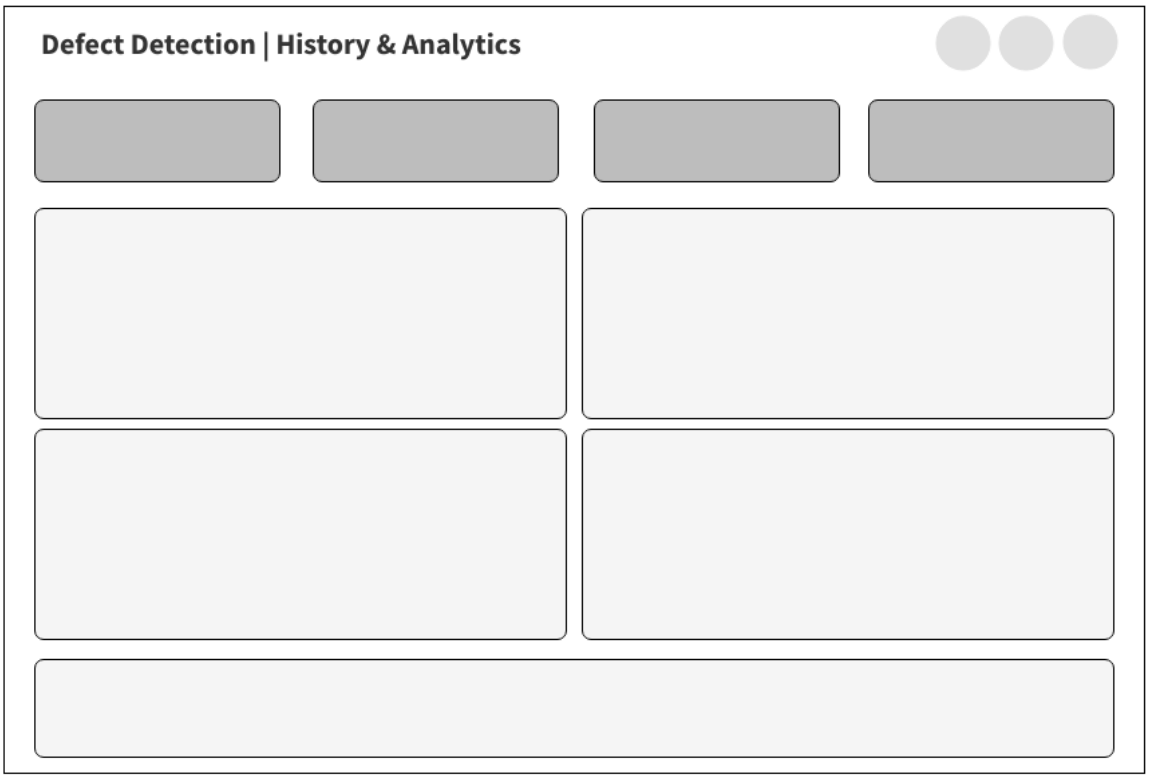


Figure 11: Component 3 Sub-Page 2 – History and Analysis Module Sketch

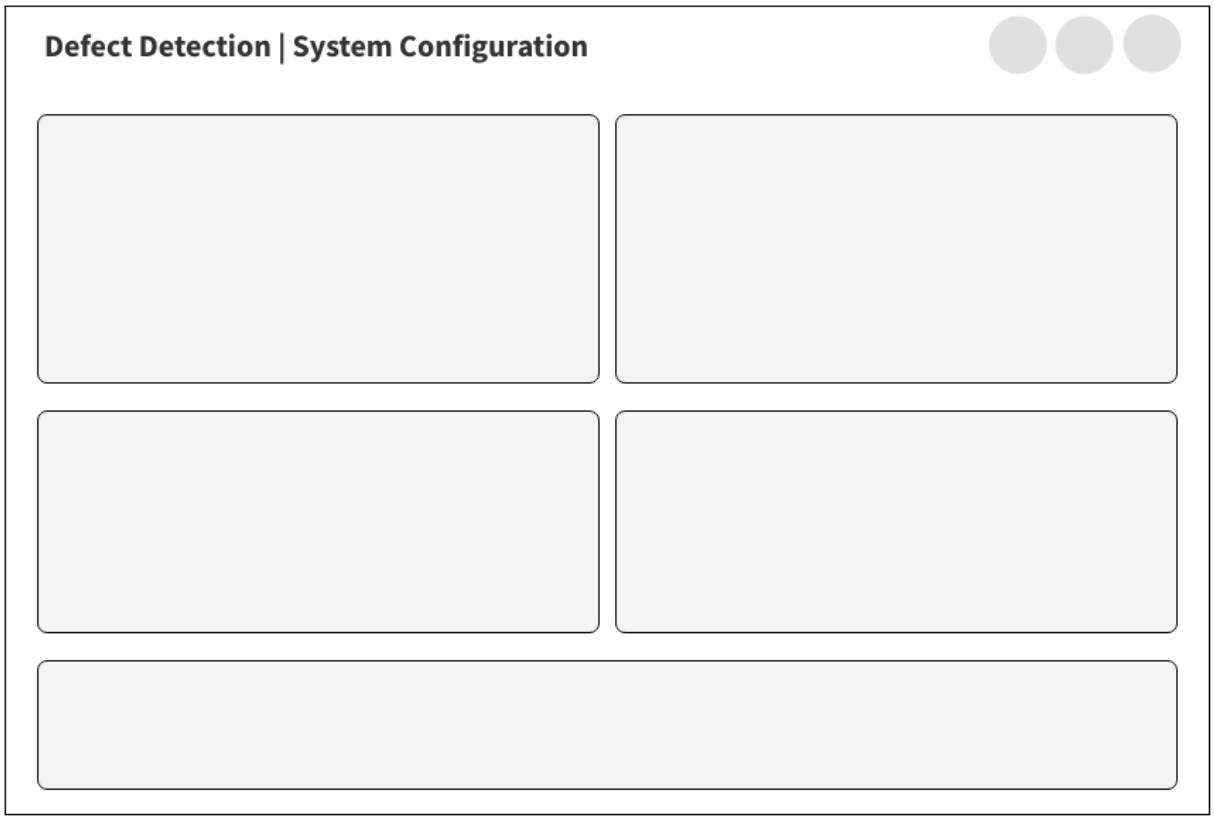
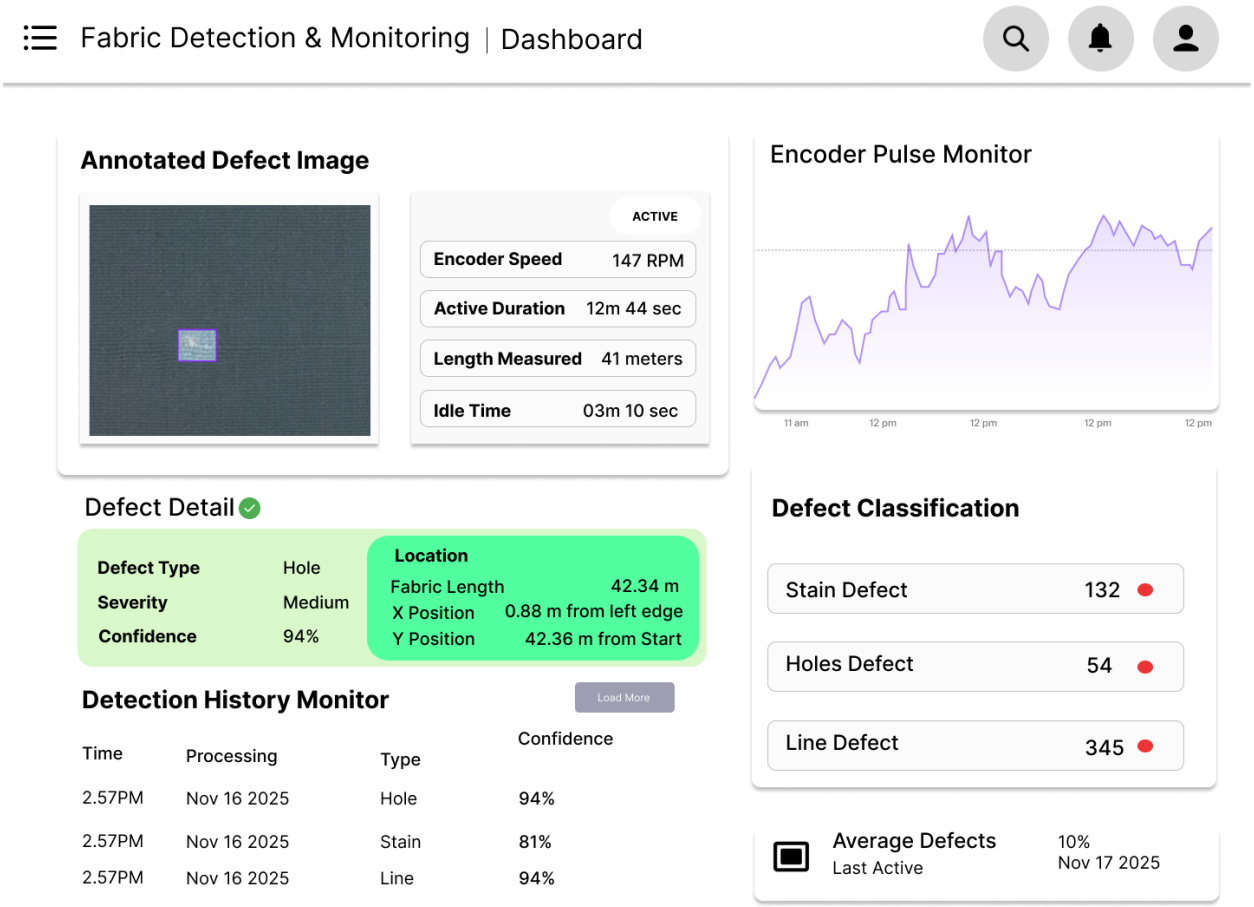
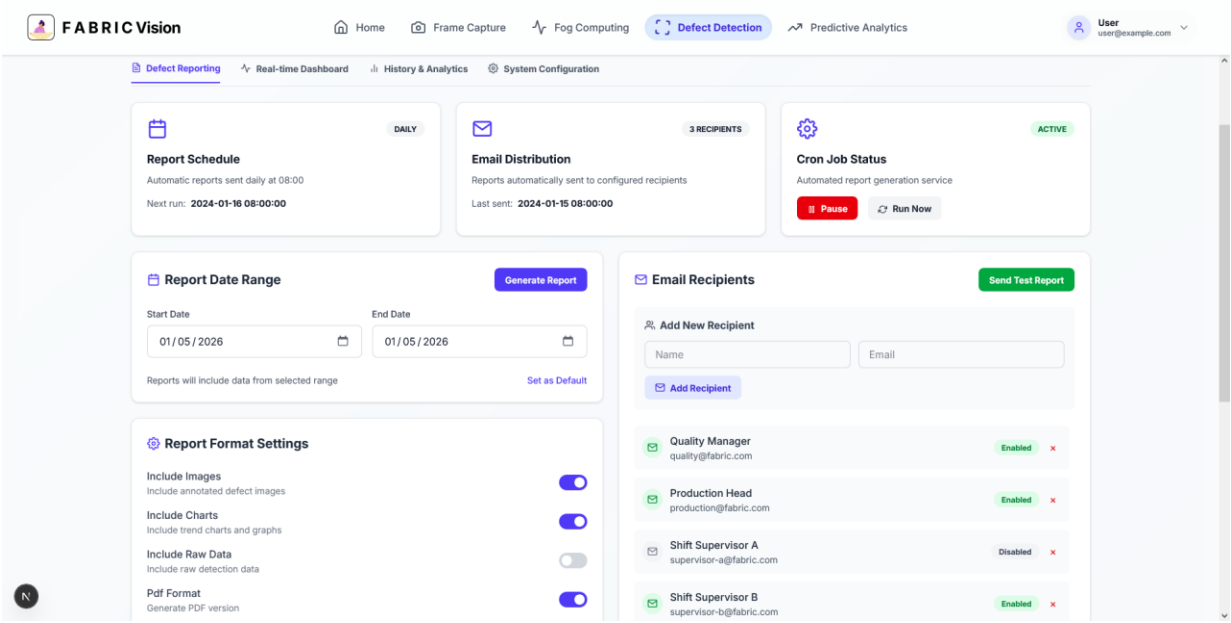


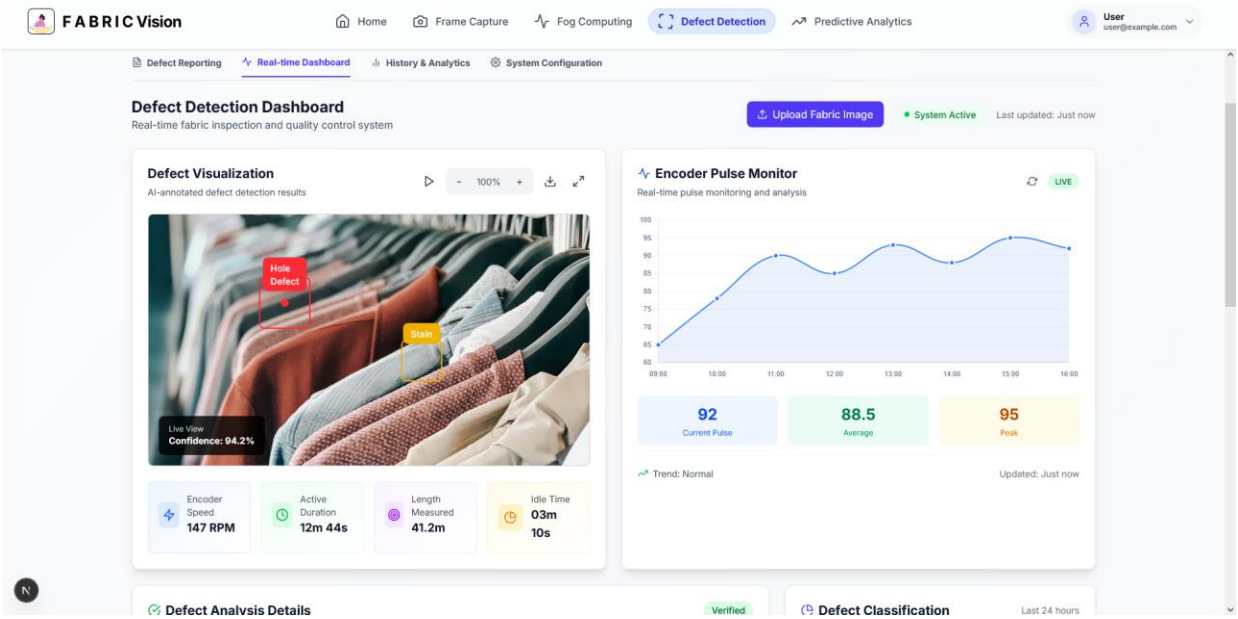
Figure 12: Component 3 Sub-Page 3 – System Configuration Module Sketch

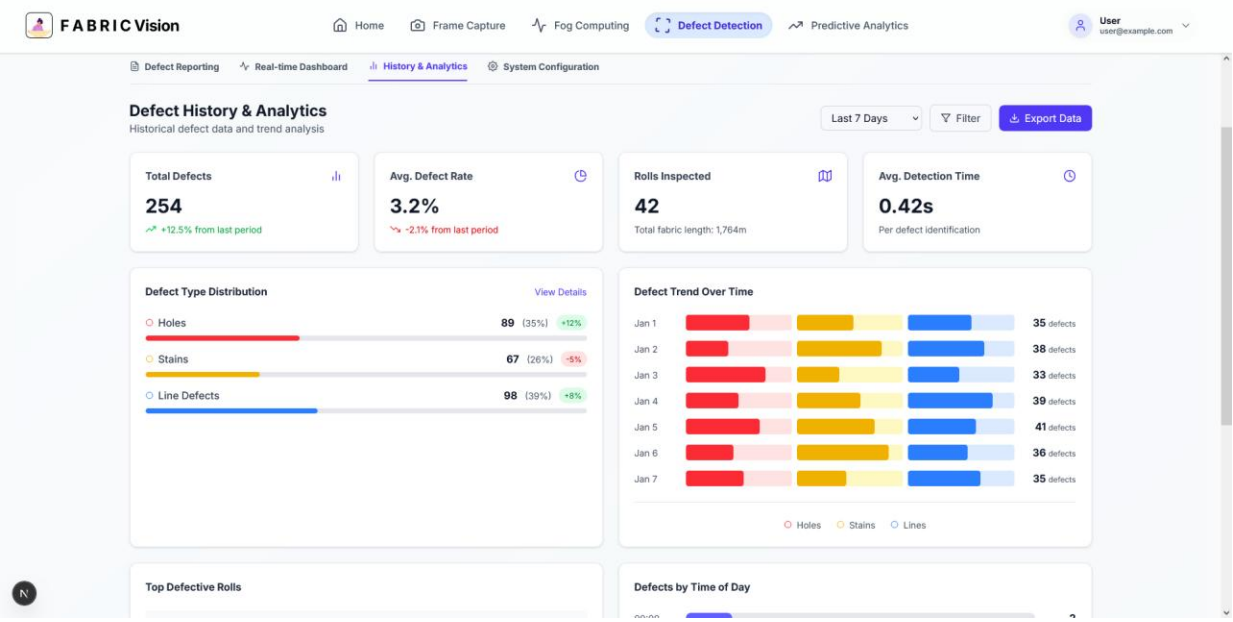
Mockups

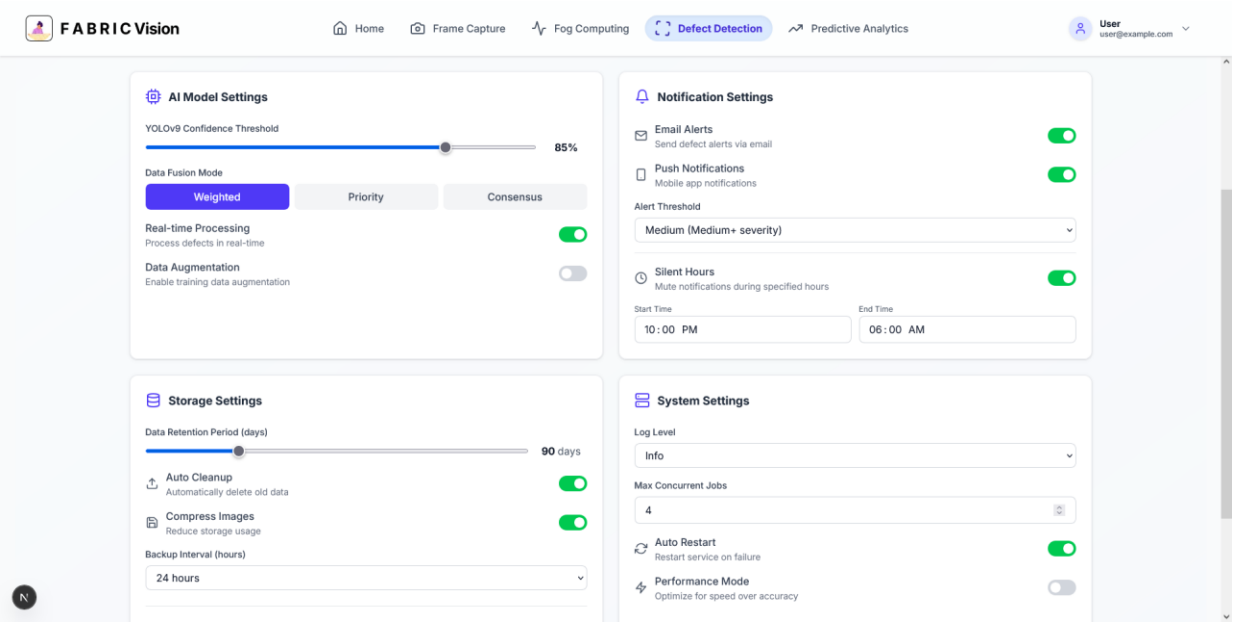

Fabric Detection & Monitoring | Dashboard
Annotated Defect Image
ACTIVE
Encoder Speed 147 RPM
Active Duration 12m 44 sec
Length Measured 41 meters
Idle Time 03m 10 sec
Encoder Pulse Monitor
11 am
12 pm
12 pm
12 pm
12 pm
Defect Detail
Defect Type Hole
Severity Medium
Confidence 94%
Location
Fabric Length 42.34 m
X Position 0.88 m from left edge
Y Position 42.36 m from Start
Defect Classification
Stain Defect 132
Holes Defect 54
Line Defect 345
Detection History Monitor
Load More
Time Processing Type Confidence
2.57PM Nov 16 2025 Hole 94%
2.57PM Nov 16 2025 Stain 81%
2.57PM Nov 16 2025 Line 94%
Average Defects 10%
Last Active Nov 17 2025

After Improvements






FABRIC Vision
Home
Frame Capture
Fog Computing
Defect Detection
Predictive Analytics
User
user@example.com
Defect Reporting
Real-time Dashboard
History & Analytics
System Configuration
Defect History & Analytics
Historical defect data and trend analysis
Last 7 Days
Filter
Export Data
Total Defects
254
+12.5% from last period
Avg. Defect Rate
3.2%
-2.1% from last period
Rolls Inspected
42
Total fabric length: 1,764m
Avg. Detection Time
0.42s
Per defect identification
Defect Type Distribution
View Details
Holes 89 (35%) +12%
Stains 67 (26%) -5%
Line Defects 98 (39%) +8%
Defect Trend Over Time
Jan 1 35 defects
Jan 2 38 defects
Jan 3 33 defects
Jan 4 39 defects
Jan 5 41 defects
Jan 6 36 defects
Jan 7 35 defects
Holes
Stains
Lines
Top Defective Rolls
Defects by Time of Day

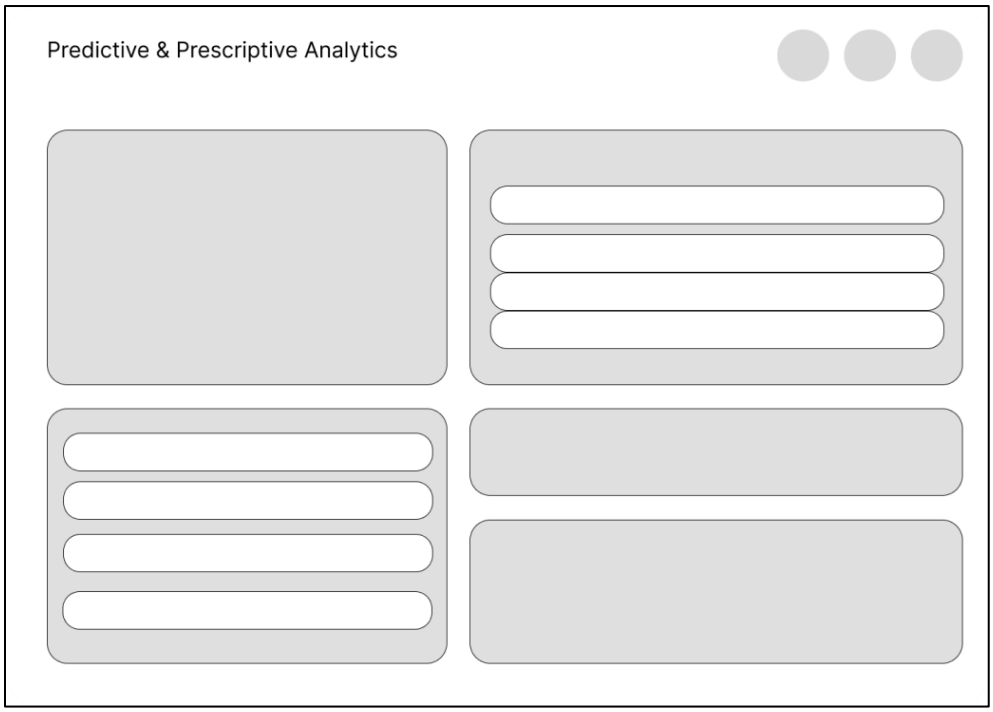

FABRIC Vision
Home
Frame Capture
Fog Computing
Defect Detection
Predictive Analytics
User
user@example.com
AI Model Settings
YOLOv9 Confidence Threshold
85%
Data Fusion Mode
Weighted
Priority
Consensus
Real-time Processing
Process defects in real-time
Data Augmentation
Enable training data augmentation
Notification Settings
Email Alerts
Send defect alerts via email
Push Notifications
Mobile app notifications
Alert Threshold
Medium (Medium+ severity)
Silent Hours
Mute notifications during specified hours
Start Time
10:00 PM
End Time
06:00 AM
Storage Settings
Data Retention Period (days)
90 days
Auto Cleanup
Automatically delete old data
Compress Images
Reduce storage usage
Backup Interval (hours)
24 hours
System Settings
Log Level
Info
Max Concurrent Jobs
4
Auto Restart
Restart service on failure
Performance Mode
Optimize for speed over accuracy

Component 4 -

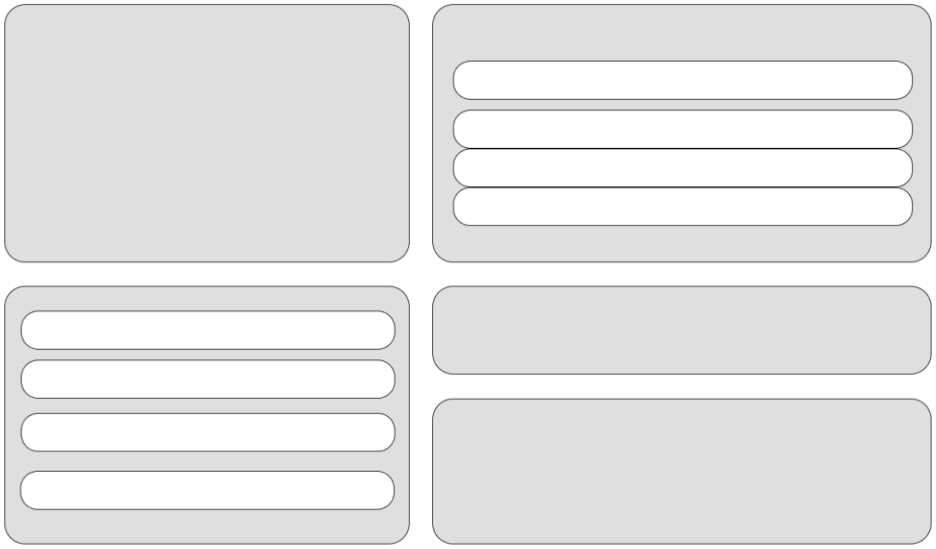
REAL-TIME FUTURE FABRIC DEFECTPREDCTION & PRESCRIPTIVEANALYTICS MODULE

FOR FUTUREFABRIC DEFECT RISK ASSESSMENT

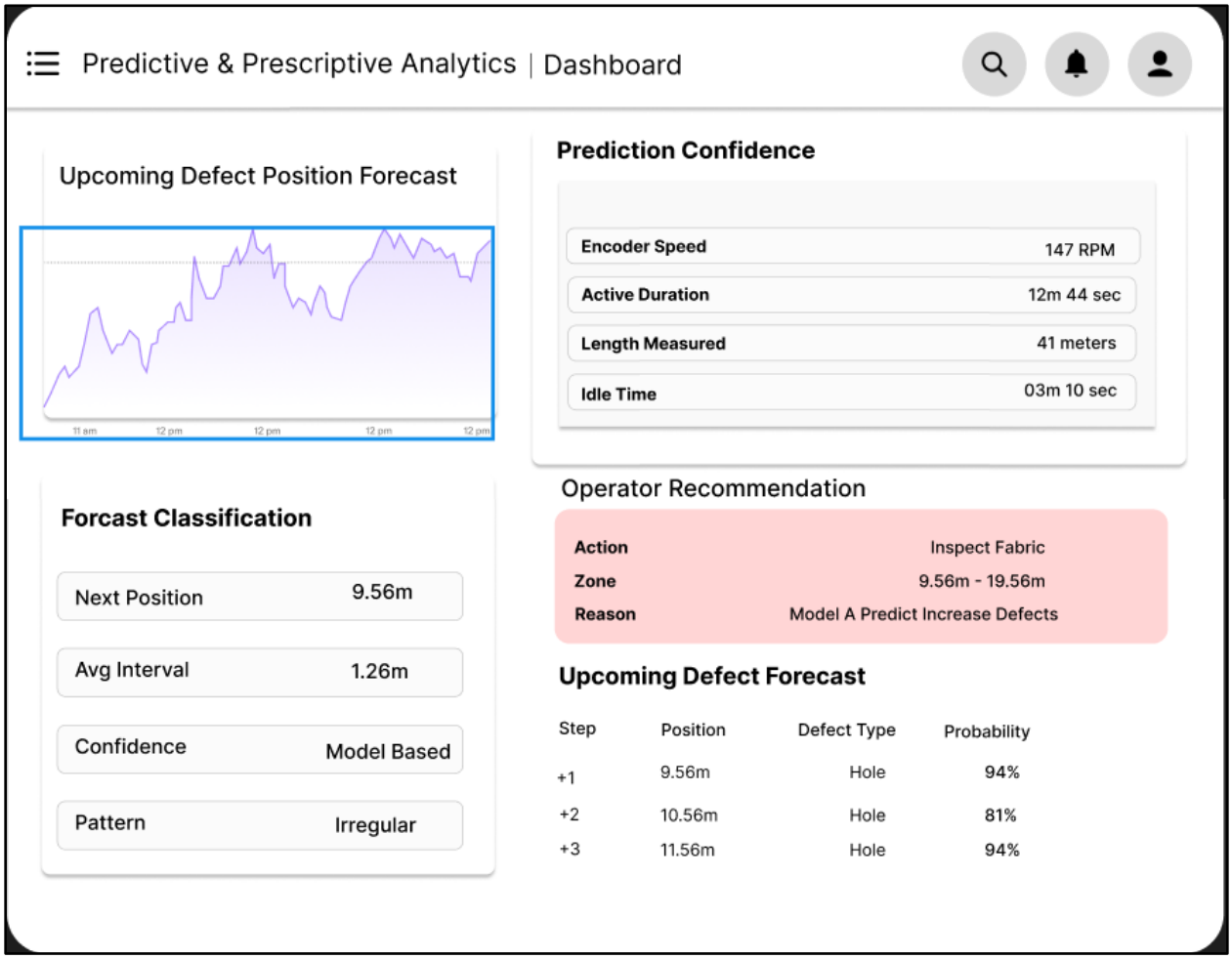
Sketches

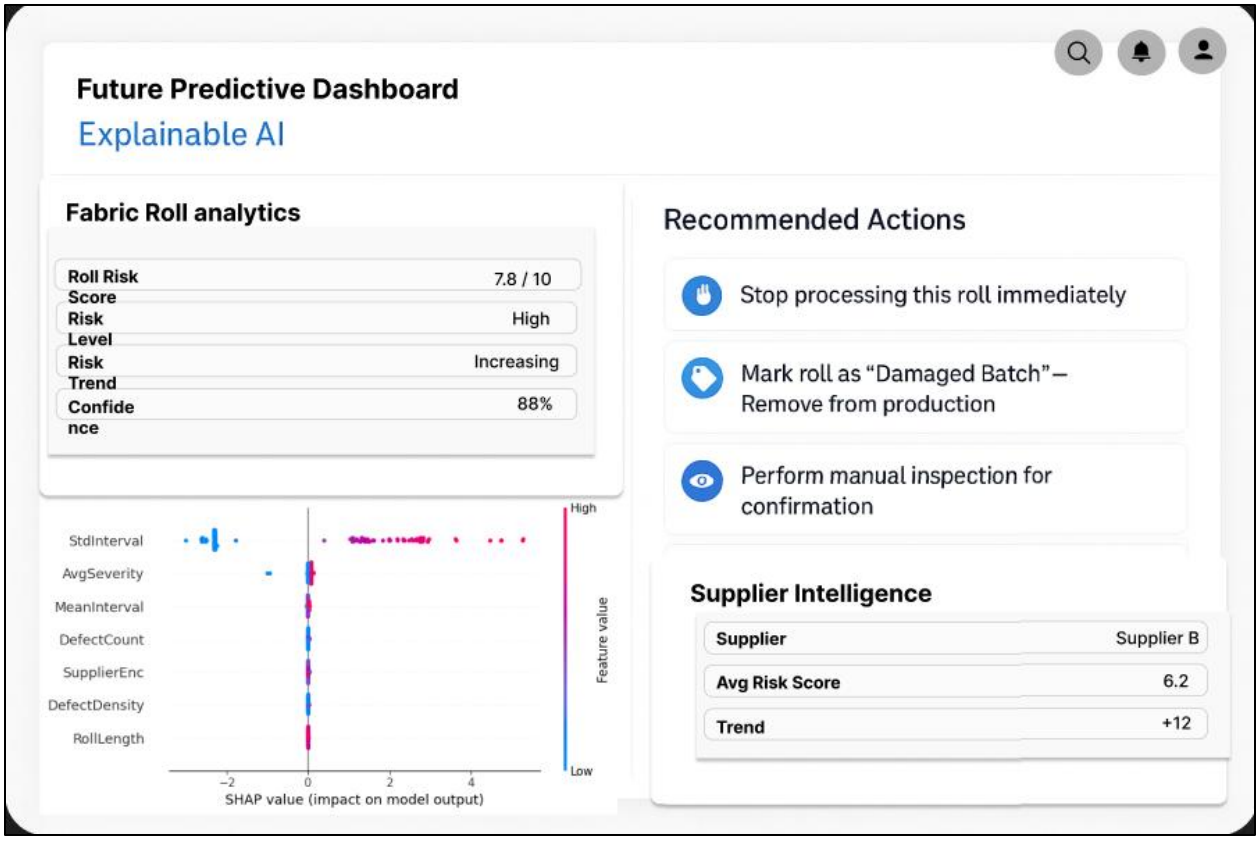


Predictive & Prescriptive Analytics

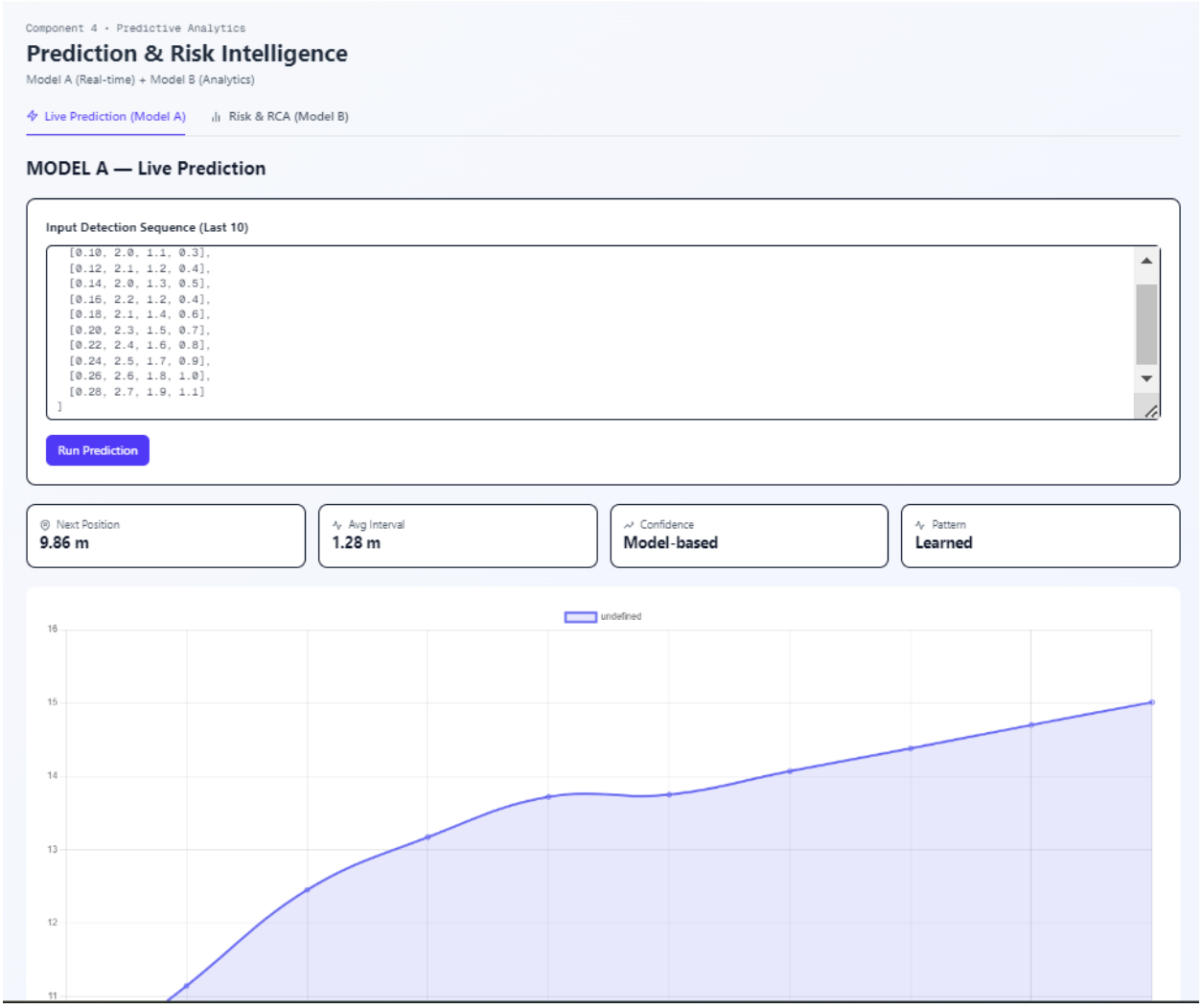


Mockups


Predictive & Prescriptive Analytics | Dashboard
Upcoming Defect Position Forecast
11 am
12 pm
12 pm
12 pm
12 pm
Prediction Confidence
Encoder Speed
147 RPM
Active Duration
12m 44 sec
Length Measured
41 meters
Idle Time
03m 10 sec
Forcast Classification
Next Position
9.56m
Avg Interval
1.26m
Confidence
Model Based
Pattern
Irregular
Operator Recommendation
Action
Inspect Fabric
Zone
9.56m - 19.56m
Reason
Model A Predict Increase Defects
Upcoming Defect Forecast
Step
Position
Defect Type
Probability
+1
9.56m
Hole
94%
+2
10.56m
Hole
81%
+3
11.56m
Hole
94%


Future Predictive Dashboard
Explainable AI
Fabric Roll analytics
Roll Risk Score 7.8 / 10
Risk Level High
Risk Trend Increasing
Confidence 88%
Recommended Actions
Stop processing this roll immediately
Mark roll as "Damaged Batch"– Remove from production
Perform manual inspection for confirmation
StdInterval
AvgSeverity
MeanInterval
DefectCount
SupplierEnc
DefectDensity
RollLength
−2
0
2
4
SHAP value (impact on model output)
High
Feature value
Low
Supplier Intelligence
Supplier Supplier B
Avg Risk Score 6.2
Trend +12

After Improvements



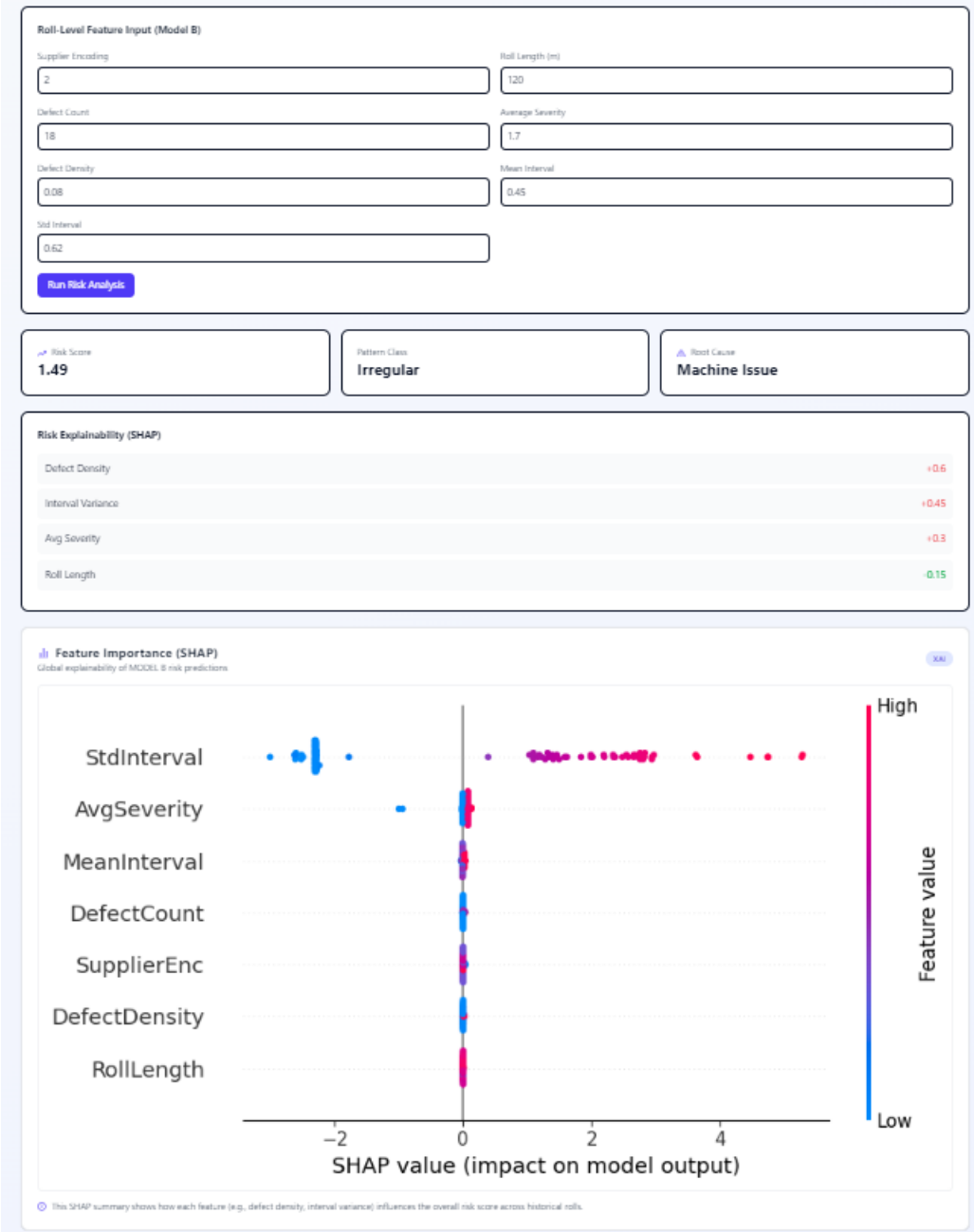
Next 10 Defect Positions

+1	9.86 m	Inspect
+2	11.14 m	Inspect
+3	12.45 m	Monitor
+4	13.17 m	Monitor
+5	13.72 m	Monitor
+6	13.75 m	Monitor
+7	14.07 m	Monitor
+8	14.38 m	Monitor
+9	14.7 m	Monitor
+10	15.01 m	Monitor

Prediction Confidence
Variance: ±0.71 m
Stability: Drifting

Pattern Evolution
Current Pattern: Drifting
Change Risk: Medium

Operator Recommendation
Action: Inspect Fabric
Zone: 9.86 – 12.45 m
Reason: Model A predicts imminent defect


Roll-Level Feature Input (Model B)
Supplier Encoding
2
Roll Length (m)
120
Defect Count
18
Average Severity
1.7
Defect Density
0.08
Mean Interval
0.45
Std Interval
0.62
Run Risk Analysis
Risk Score
1.49
Pattern Class
Irregular
Root Cause
Machine Issue
Risk Explainability (SHAP)
Defect Density
+0.6
Interval Variance
+0.45
Avg Severity
+0.3
Roll Length
-0.15
Feature Importance (SHAP)
Global explainability of MODEL B risk predictions
XAI
StdInterval
AvgSeverity
MeanInterval
DefectCount
SupplierEnc
DefectDensity
RollLength
−2
0
2
4
SHAP value (impact on model output)
High
Feature value
Low
This SHAP summary shows how each feature (e.g., defect density, interval variance) influences the overall risk score across historical rolls.